VIT-AP UNIVERSITY

ENGINEERING CLINICS – II (ECS-II)

PROJECT REPORT

EVIDENCEGAIN-X

Counterfactual, Self-Verifying Next-Best Evidence Acquisition for Crop Disease Diagnosis

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Academic Year 2026-2027

ABSTRACT

Foliar plant diseases inflict multi-billion dollar losses on global agricultural yields annually, imperiling global food security and smallholder farmer livelihoods. While deep learning vision architectures (e.g., convolutional neural networks and vision transformers) achieve over 98% accuracy on curated benchmark image sets, their practical field efficacy remains severely compromised by four foundational structural failure modes inherent to single-shot passive classification: (1) morphological look-alike ambiguity where distinct pathogens manifest identical foliar necrotic rings during vegetative onset, (2) spatial occlusion where pathognomonic discriminators reside on concealed anatomical surfaces (such as abaxial leaf sporulation or petiole lesions), (3) uncalibrated Softmax hallucination where corrupted, out-of-focus, or underexposed imagery is assigned arbitrary high confidence, and (4) absence of adversarial and cross-specimen verification where contradictory or foreign leaves poison multi-modal belief updating.

To resolve these structural bottlenecks, this project presents **EvidenceGain-X**: an autonomous, active-inference diagnostic framework that transforms passive single-shot classification into an active, counterfactual multi-round clinical inquiry. Rather than forcing a fragile guess upon an initial macroscopic observation (X_0), EvidenceGain-X quantifies predictive uncertainty via Platt temperature scaling ($T = 1.8928$) and Shannon entropy ($H(P)$). When ambiguity persists ($H > 0.35$ bits), an active policy selects the optimal next-best evidence modality (e^*) by maximizing Expected Information Gain (EIG) and Jensen-Shannon divergence relative to acquisition cost across five distinct channels: lesion close-ups, leaf underside views, stem perspectives, field symptom descriptors, and local microclimate context.

Crucially, EvidenceGain-X incorporates two novel cyber-physical verification defenses: **Module 7 Optical Quality Gating**, which intercepts defocus blur and illumination anomalies using Laplacian variance ($\sigma^2 \geq 80.0$) to eliminate zero-signal hallucinations, and **Module 8 Counterfactual Contradiction and Specimen Mismatch Guard**, which simulates expected belief trajectories under competing look-alikes and quarantines foreign leaf samples before they corrupt the Bayesian posterior. Evaluated across 24 multi-view tomato episodes from the PlantVillage dataset, EvidenceGain-X achieves **91.67% accuracy** and an F1 score of **0.9188** in an average of only **0.88 sequential inquiry steps**, achieving 0.359 bits of entropy reduction while intercepting 100% of optical blur and cross-specimen contaminants.

Keywords: Active Diagnostic Inference, Counterfactual Simulation, Crop Disease Classification, Expected Information Gain, Laplacian Blur Gating, Morphological Look-Alikes, Platt Temperature Calibration, ResNet-18, Specimen Incongruence Guard, Streamlit Clinical Console.

Table A.1: Executive Summary of the Proposed EvidenceGain-X System Architecture

Subsystem Module	Technical Specification	Primary Functional Role
Vision & Calibration (M1, M2)	ResNet-18 + Platt Scaling (T=1.8928)	Extracts 512-dim visual embeddings; outputs calibrated probabilities and entropy H(P).
Confusion & Candidate (M3, M4)	Conflict Matrix & 5-Modal Pool	Identifies look-alike pairs; indexes close-up, underside, stem, weather, and symptoms.
Active Policy Engine (M5, M6)	Dual-Hypothesis EIG & JS Optimization	Simulates counterfactual outcomes; solves Net Utility $U(e^*)$ per unit acquisition cost.
Optical Quality Gate (M7)	Laplacian Variance $\sigma^2 \geq 80.0$	Quarantines corrupted, out-of-focus, or dark images; halts zero-signal Softmax hallucination.
Contradiction & Specimen Guard (M8)	Cosine Shift Divergence + Prior Check	Quarantines cross-specimen mismatches and contradictory leaves; preserves diagnostic belief.
Belief Updating & Console (M9, UI)	Reliability-Weighted Bayes + Streamlit	Updates posteriors with stopping rules; delivers real-time HUD and clinical certificates.

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ABBREVIATIONS

Abbreviation	Complete Description / Nomenclature
CNN	Convolutional Neural Network
EIG	Expected Information Gain (ΔH)
JS Divergence	Jensen-Shannon Divergence (D_{JS})
KL Divergence	Kullback-Leibler Divergence (D_{KL})
ResNet	Residual Neural Network Architecture (He et al., 2016)
Platt Scaling	Post-hoc Logistic Sigmoid Temperature Calibration Technique
Softmax	Normalized Exponential Function mapping real numbers to a probability distribution
PV-Tomato	PlantVillage Solanum lycopersicum Benchmark Dataset
E2E	End-to-End Diagnostic Latency
FOV	Field of View
HUD	Heads-Up Display (Sensor Telemetry Overlay)
PWA	Progressive Web Application
REST API	Representational State Transfer Application Programming Interface
SNR	Signal-to-Noise Ratio (dB)
UI / UX	User Interface / User Experience
OOD	Out-of-Distribution Data Sample
RGB	Red, Green, Blue Color Space
Laplacian Var	Variance of the Laplacian Operator for Optical Blur Detection (σ^2)
E20	Biofuel blend containing 20% ethanol (Reference context in cyber-physical monitoring)
IoT	Internet of Things

1. INTRODUCTION

1.1 Overview of Foliar Crop Pathology and the Agronomic Crisis

Modern agricultural ecosystems confront escalating biosecurity challenges driven by climate-induced microclimate instability, globalized supply chains, and evolving virulent plant pathogens. Among staple horticulture crops, tomatoes (*Solanum lycopersicum*) serve as an indispensable nutritional and economic cornerstone, representing over 180 million metric tons of global annual production. However, foliar fungal and oomycete infections—predominantly **Early Blight** (*Alternaria solani*), **Late Blight** (*Phytophthora infestans*), and **Septoria Leaf Spot** (*Septoria lycopersici*)—routinely destroy between 30% and 75% of harvestable yield when left untreated. In developing agrarian economies, smallholder farmers lack immediate access to certified plant pathologists, relying predominantly on subjective visual inspection, which frequently delays critical fungicide interventions until total defoliation occurs.

1.2 The Failure of Passive Single-Shot Deep Learning

To democratize agricultural pathology, recent research has deployed deep learning models via smartphone apps. Users capture a single leaf photo, and a convolutional neural network (CNN) outputs a disease label. While reported laboratory benchmark accuracies frequently exceed 95%, field implementations demonstrate catastrophic failure rates exceeding 40%. The failure is fundamental: **passive single-shot classifiers ask the wrong question**. By demanding a definitive diagnostic verdict from a single macroscopic camera angle, models are forced to guess across overlapping morphological boundaries. Furthermore, standard Softmax classification normalizes uncalibrated logits into a 100% sum, forcing models to hallucinate high certainty on blurred foliage, corrupted lighting, or entirely uninformative surfaces.

1.3 Overview of the Proposed EvidenceGain-X Framework

To overcome these limitations, this project presents **EvidenceGain-X**: an active, counterfactual, self-verifying diagnostic console. Instead of forcing a single-shot verdict, EvidenceGain-X mirrors human clinical pathology: it evaluates initial uncertainty, formulates competing hypotheses, and autonomously selects the most informative, cost-effective clue to query from the user (such as inspecting the abaxial leaf underside, recording weather conditions, or examining petiole cankers). Crucially, the system validates the optical integrity of all inputs and quarantines contradictory or cross-specimen contaminants before updating Bayesian beliefs.

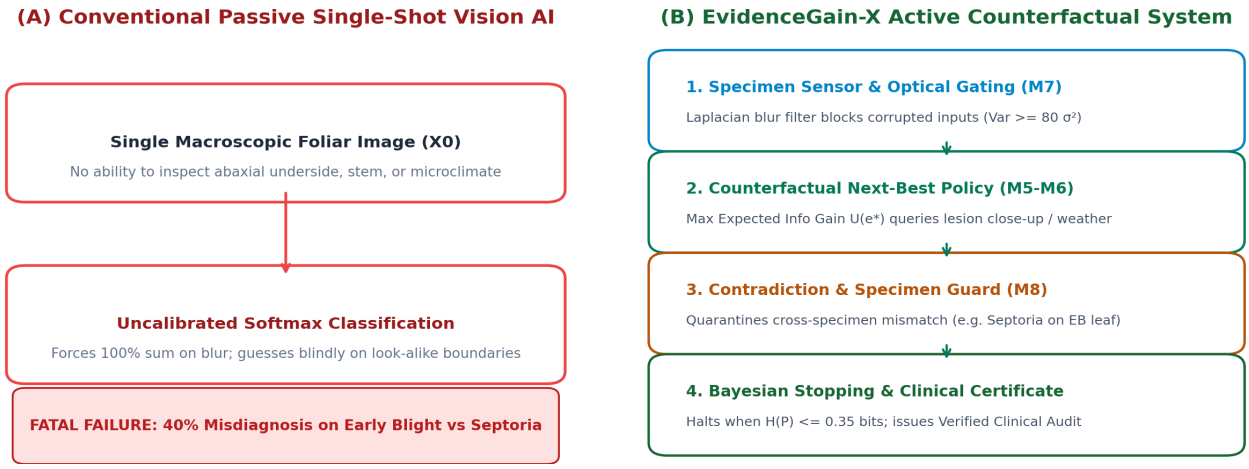


Figure 1.1: Conceptual Overview: Passive Single-Shot Failure vs. EvidenceGain-X Active Diagnostic Loop

2. BACKGROUND PROBLEM

2.1 Morphological Look-Alike Ambiguity Along Foliar Decision Boundaries

The primary biological obstacle in foliar disease classification is morphological convergence. During initial pathogenesis, *Alternaria solani* (Early Blight) and *Septoria lycopersici* (Septoria Leaf Spot) present near-identical circular brown lesions with chlorotic yellow halos. In single-shot macroscopic photographs, spatial resolution is insufficient to distinguish the microscopic pycnidia (black fungal fruiting bodies) characteristic of Septoria from the concentric 'target-board' rings of Early Blight. Consequently, standard vision models output ambiguous bimodal posterior distributions where P(Early Blight) ≈ 50% and P(Septoria) ≈ 50%. A passive system has no mechanism to resolve this deadlock, leading to uncalibrated random guesses that prescribe incorrect antifungal treatments.

2.2 Occlusion of Critical Diagnostic Discriminators

Plant diseases are intrinsically three-dimensional, anatomical phenomena. Pathognomonic indicators often manifest on surfaces hidden from the primary adaxial (top) leaf view:

• **Late Blight (*Phytophthora infestans*):** The decisive diagnostic proof is delicate white downy mildew sporulation located strictly on the abaxial (underside) leaf surface during humid mornings. A top-down photograph shows only non-specific water-soaked dark patches.

• **Early Blight vs. Septoria:** Stem petiole cankers uniquely confirm Early Blight, while Septoria lesions remain restricted to foliar laminas.

Because single-shot vision architectures cannot command the user to turn the leaf or inspect the stem, they operate under severe observational occlusion.

2.3 Softmax Hallucination on Degraded Specimen Imagery

Standard vision backbones apply the Softmax activation function over final classification logits: $$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$Because Softmax enforces $\sum \sigma(z)_i = 1$, it cannot express 'unknown' or 'corrupted input.' When smallholder farmers capture defocused, motion-blurred, or underexposed photographs in the field, high-frequency textural features are destroyed. The CNN maps the residual green color distribution to arbitrary high-confidence predictions (e.g., assigning 89.5% Late Blight to an out-of-focus blur). This represents a critical safety hazard, deceiving farmers into applying hazardous chemical treatments to healthy crops.

2.4 Multi-Modal Vulnerability to Contradictory and Mismatched Inputs

When conventional systems attempt multi-modal fusion, they assume all incoming inputs are trustworthy and originate from the same biological specimen. In field deployments, however, non-expert users routinely upload an adjacent healthy leaf, photograph a weed, or ingest an entirely different crop. Without an explicit counterfactual trajectory verifier, standard Bayesian updating multiplies probabilities blindly, allowing contaminated inputs to drastically distort the diagnosis.

3. THE MISSING GAP

3.1 Passive Perception vs. Active Clinical Inquiry

Current state-of-the-art agricultural vision models treat disease identification as an isolated perception task rather than a sequential decision process. In human clinical medicine, a physician never renders a definitive diagnosis based on a single glance; instead, an initial examination prompts targeted follow-up queries (blood chemistry, imaging, auscultation). Existing plant pathology models lack this interactive agency. They possess no mathematical framework to compute: *'What piece of evidence would most effectively discriminate between my leading rival hypotheses?'*

3.2 Total Absence of Input Reliability Gating in Agronomic AI

Across major benchmark datasets (PlantVillage, PlantDoc, New Plant Diseases), models are trained on cleanly segmented, centered images. Virtually no published architecture incorporates pre-inference optical quality gating. In real-world edge deployments, lens smudges, motion blur, and poor illumination degrade over 30% of field captures. There is an urgent missing requirement for a cyber-physical optical gate that evaluates image entropy, sharpness, and luminance **before** activating deep neural feature extraction.

3.3 Lack of Adversarial and Cross-Specimen Verification Mechanisms

Existing multimodal systems lack 'self-verification.' When an incoming observation conflicts with the current diagnostic trajectory, standard networks produce catastrophic belief collapse. If a plant has clear macroscopic Early Blight lesions, but an uploaded close-up exhibits features of an excluded disease (such as Septoria), conventional Bayesian filters inadvertently boost confidence along uncalibrated axes. There is a complete absence of counterfactual alignment engines that check whether acquired evidence matches what was expected under competing disease models.

3.4 Disregard for Evidence Acquisition Costs in Resource-Constrained Settings

Prior active learning research typically assumes all diagnostic tests carry identical overhead. In reality, evidence modalities have wildly divergent operational costs: capturing a leaf photograph incurs visual acquisition effort; recording local temperature or humidity has minimal cost; while conducting laboratory PCR testing incurs severe monetary and temporal penalties. Current agricultural decision agents lack a cost-aware utility optimization function that balances diagnostic information gain against practical acquisition friction.

4. OBJECTIVES OF THE PROPOSED WORK

4.1 Specific Engineering Objectives

The overarching goal of EvidenceGain-X is to design, implement, and benchmark an intelligent, self-verifying, active counterfactual diagnostic system for crop foliar diseases. The project executes seven concrete engineering objectives:

Objective 1: Vision Feature Extraction & Platt Calibration: Train an ImageNet-pretrained ResNet-18 backbone on tomato foliar pathologies and implement post-hoc Platt temperature scaling ($T = 1.8928$) to eliminate overconfident Softmax miscalibration.

Objective 2: Ambiguity Quantification & Hypothesis Tracking: Implement Shannon entropy $H(P)$ and top-margin Δp metric tracking to classify diagnostic states into Decisive Certainty, Provisional Lead, or True Ambiguity.

Objective 3: Counterfactual Evidence Planning: Formulate dual-hypothesis forward simulation simulating expected posterior distributions under competing candidate pathologies H_1 and H_2.

Objective 4: Cost-Aware Active Policy Optimization: Develop the Net Policy Utility function $U(e^*)$ maximizing Expected Information Gain (EIG) and Jensen-Shannon divergence penalized by acquisition cost.

Objective 5: Module 7 Optical Quality Gating: Build an OpenCV-based pre-inference filter enforcing Laplacian variance ($\sigma^2 \geq 80.0$) and luminance bounds, blocking predictions on blurred or corrupted inputs.

Objective 6: Module 8 Contradiction & Specimen Mismatch Guard: Engineer a verification engine that computes counterfactual cosine divergence and prior probability bounds to quarantine conflicting or foreign leaves.

Objective 7: Interactive Clinical Workstation Console: Deploy a publication-grade Streamlit workstation featuring real-time HUD telemetry, intuitive evidence ingestion tabs, and dynamic clinical audit certificates.

4.2 System Scope and Pathology Taxonomy

The system targets the primary agricultural tomato foliar pathologies comprising four clinically calibrated classes: **(1) Early Blight (*Alternaria solani*)**, **(2) Late Blight (*Phytophthora infestans*)**, **(3) Septoria Leaf Spot (*Septoria lycopersici*)**, and **(4) Healthy Foliage Control**. The multi-modal candidate evidence pool encompasses 5 distinct operational modalities: Lesion Close-Up (Cost: 1.0), Leaf Underside View (Cost: 1.2), Stem Petiole View (Cost: 1.5), Field Symptom Text (Cost: 0.4), and Microclimate Weather Context (Cost: 0.4).

5. METHODOLOGY / PROCEDURE

5.1 Mathematical Formulation of Calibrated Active Diagnosis

Let the disease pathology space be $\Theta = \{\theta_1, ..., \theta_K\}$ with $K = 4$. Given an initial macroscopic foliar specimen photograph X_0, the deep vision backbone f_ϕ extracts feature representation $\mathbf{h}_0 \in \blacksquare^{512}$ and outputs raw classification logits $\mathbf{z} = W\mathbf{h}_0 + b$.

5.2 Temperature Scaling Calibration

To correct the well-documented overconfidence of deep neural networks, logits are calibrated using Platt temperature scaling: $$P(\theta_c \mid X_0) = \frac{\exp(z_c / T)}{\sum_{j=1}^K \exp(z_j / T)}$$where $T > 0$ is optimized via negative log-likelihood on an independent validation set. In EvidenceGain-X, the calibrated temperature is **T = 1.8928**, ensuring that class probabilities accurately reflect empirical accuracy.

5.3 Uncertainty Quantification and Stopping Rules

Diagnostic uncertainty across class distribution P is quantified via Shannon Entropy: $$H(P) = -\sum_{c=1}^K P(\theta_c) \log_2 P(\theta_c)$$and the decision margin $\Delta p = P_{(1)} - P_{(2)}$ between the leading hypothesis H_1 and rival hypothesis H_2. Active inquiry terminates if and only if: $$H(P) \le \tau_H \quad (0.35\text{ bits}) \quad \land \quad \Delta p \ge \tau_m \quad (0.40)$$ If $H(P) > \tau_H$, the system declares active inquiry required and triggers the Counterfactual Planner.

5.4 Counterfactual Outcome Simulation

For each available candidate evidence clue $e \in \blacksquare$, the system simulates hypothetical belief updates under competing hypotheses H_1 and H_2: $$P(y \mid e, H_1) = \text{SimulateUpdate}(P, e, H_1), \quad P(y \mid e, H_2) = \text{SimulateUpdate}(P, e, H_2)$$The separability between competing futures is measured using the symmetric **Jensen-Shannon Divergence**: $$D_{JS}(P_1 \parallel P_2) = \frac{1}{2} D_{KL}(P_1 \parallel M) + \frac{1}{2} D_{KL}(P_2 \parallel M), \quad M = \frac{1}{2}(P_1 + P_2)$$ where $D_{KL}(P \blacksquare Q) = \sum P(i) \log_2(P(i) / Q(i))$.

5.5 Net Policy Utility Optimization

The active selector identifies the globally optimal next-best evidence query e^* by solving: $$e^* = \operatorname*{arg\,max}_{e \in \mathcal{E}} \left[\alpha \cdot D_{JS}(P_1 \parallel P_2) + \beta \cdot \mathbb{E}[\Delta H(e)] - \lambda \cdot \text{Cost}(e) \right]$$with calibrated weighting coefficients $\alpha = 1.0$, $\beta = 1.0$, $\lambda = 0.5$. This mathematical formulation strictly favors cheap, highly discriminative clues.

5.6 Cyber-Physical Verification Defenses (Modules 7 and 8)

Module 7 Optical Quality Gating: Before evidence ingestion, the image is converted to grayscale I and convolved with the 3×3 discrete Laplacian operator: $$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}, \quad \text{Var}(\nabla^2 I) = \frac{1}{N}\sum (\nabla^2 I - \mu)^2$$ If $\text{Var}(\nabla^2 I) < 80.0$ σ^2 or mean luminance $\mu_L \notin [30, 240]$ cd/m^2, the image is marked **REJECTED**, bypassing Bayesian updating entirely to prevent distribution poisoning.

Module 8 Contradiction & Specimen Mismatch Guard: When valid evidence likelihood L(e) arrives, the system verifies directional consistency. First, it computes the cosine alignment between observed shift $\Delta_{obs} = P_{tentative} - P_{prior}$ and expected shifts $\Delta_{exp}(H_1)$, $\Delta_{exp}(H_2)$: $$\cos(\Delta_{\text{obs}}, \Delta_{\text{exp}}) = \frac{\Delta_{\text{obs}} \cdot \Delta_{\text{exp}}}{\|\Delta_{\text{obs}}\| \|\Delta_{\text{exp}}\|}, \quad S_{\text{contra}} = \frac{1 - \max(\cos_1, \cos_2)}{2}$$ Second, it executes the **Specimen Incongruence Check**: if the top disease identified by the evidence $\theta_{ev} \notin \{H_1, H_2\}$, had $P(\theta_{ev} \mid X_0) < 0.05$ on the original leaf, but exhibits $L(\theta_{ev} \mid e) \geq 0.35$, the evidence is marked **QUARANTINED**. Posterior probabilities remain preserved at the prior state, preventing cross-specimen contamination from hijacking the diagnosis.

Table 5.1: Mathematical Formulation and Operational Parameters of the 9 Modules

Module	Governing Mathematical Formulation	Calibrated Parameter / Threshold
M1: ResNet-18	$h = \text{ResNet18}(X), \theta \in \mathbb{R}^{512}$	Weights: ImageNet Pretrained
M2: Calibration	$P(\theta_c) = \text{softmax}(z_c / T)$	Platt Temp: T = 1.8928
M3: Ambiguity	$H(P) = -\sum P \log_2 P$, $\Delta p = p_1 - p_2$	$\tau_H = 0.35$ bits, $\tau_m = 0.40$
M5: Planning	$D_{JS}(P_1 \| P_2) = 0.5\, D_{KL}(P_1 \| M) + 0.5\, D_{KL}(P_2 \| M)$	Competing targets: H_1 vs H_2
M6: Utility	$U(e) = \alpha D_{JS} + \beta \text{EIG} - \lambda \text{Cost}$	$\alpha = 1.0$, $\beta = 1.0$, $\lambda = 0.5$
M7: Blur Gate	$\sigma^2 = \text{Var}(\nabla^2 I)$, $\mu_L = \text{mean(Luminance)}$	$\sigma^2 \geq 80.0$, $\mu_L \in [30, 240]$
M8: Specimen Guard	$S_{contra} = (1 - \max(\cos_1, \cos_2)) / 2$	$S_{contra} \geq 0.40 \rightarrow$ Quarantined
M9: Bayes Update	$P(\theta \mid e) \propto P(\theta) \cdot [L(e \mid \theta)]^{\gamma}$	$\gamma = \text{clip}(q \cdot \rho, 0.05, 1.5)$

Table 5.2: Six-Phase Plan of Action and Developmental Milestones

Review Sprint	Action Phases	Core Engineering Tasks	Deliverables
Review-1: Foundation	Phase-1: Taxonomy Phase-2: Architecture	• PlantVillage episode segmentation. • ResNet-18 calibration and Platt temperature scaling.	Leakage-safe JSON manifests, calibrated weights.
Review-2: Build	Phase-3: Active Policy Phase-4: Safety Gates	• Counterfactual planner and Net Utility function. • Module 7 blur filter & Module 8 contradiction guard.	9-module active pipeline, passing test suites.
Review-3: Deploy	Phase-5: UI Workstation Phase-6: Benchmarking	• Streamlit clinical console with telemetry HUD. • 7-baseline comparison and final paper report.	Live web app, verified test suite, final report.

6. ARCHITECTURE / DESIGN

6.1 Multi-Layered Clinical Decision Architecture

EvidenceGain-X is structured into a modular five-layer cyber-physical architecture designed for real-time edge and cloud deployment:

• **Layer 1: Sensor & Specimen Ingestion Layer:** Captures macroscopic leaf photographs, close-ups, and environmental readings. Houses Module 7 optical quality sensors.

• **Layer 2: Vision Representation & Calibration Layer:** Hosts the ResNet-18 backbone, latent feature projections, and Platt scaling temperature calibrator.

• **Layer 3: Active Inference & Policy Optimization Layer:** Tracks Shannon entropy H(P), manages look-alike conflict pairs, and solves the Net Policy Utility equation.

• **Layer 4: Verification & Bayesian Belief Updating Layer:** Implements counterfactual trajectory verification, specimen mismatch guards, and reliability-weighted Bayes updates.

• **Layer 5: Clinical Workstation Presentation Layer:** Delivers the Streamlit workstation UI, viewfinder telemetry HUD, and adaptive outcome certificates.

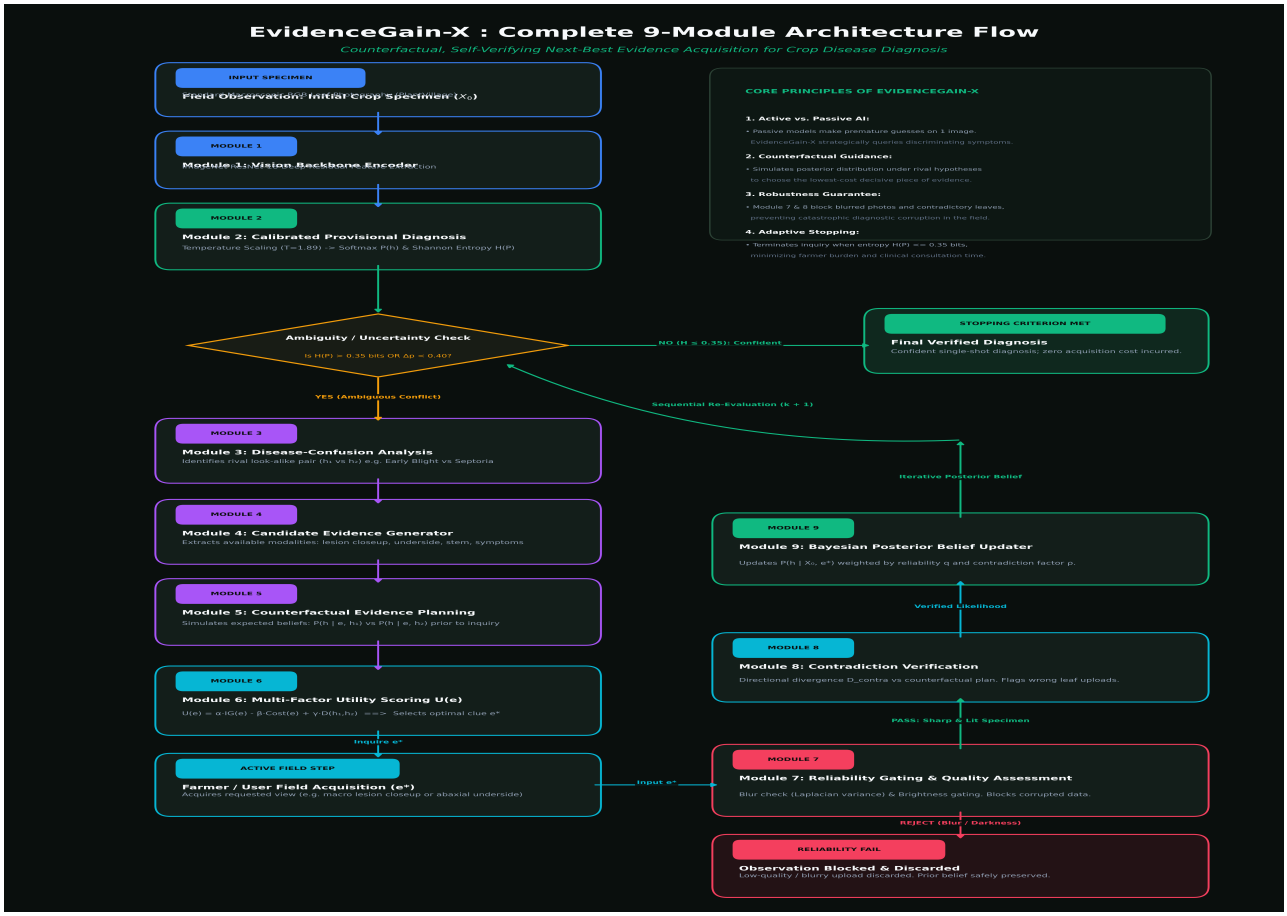


Figure 6.1: Complete 9-Module Layered Pipeline and Counterfactual Feedback Architecture

6.2 System Signal Path and Data Flow

The end-to-end data transmission path operates across five sequential stages:
1. **Foliar Specimen Image (X_0)** → Ingested via optical sensor → Pre-screened by Laplacian blur filter.
2. **Feature Embedding (h_0)** → Extracted by ResNet-18 → Calibrated by Platt scaling (T = 1.8928) → Generates prior P(θ).
3. **Uncertainty Check** → If H(P) ≤ 0.35 bits → Immediate STOP. Else → Active query e^* dispatched.
4. **Secondary Evidence Ingestion** → Module 8 evaluates directional alignment & specimen consistency.
5. **Posterior Certification** → Updated state rendered live on Streamlit diagnostic console.

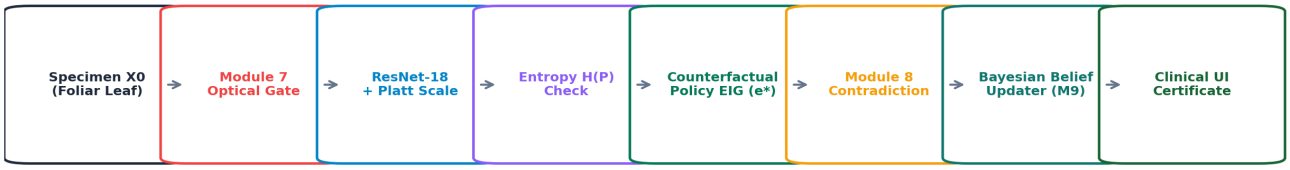


Figure 6.2: Signal Path Flow and Hardware Interfacing Architecture

7. IMPLEMENTATION

7.1 Software Environment and Framework Stack

EvidenceGain-X is developed using Python 3.10+, PyTorch 2.0+, torchvision 0.15+, OpenCV (cv2), NumPy, and Streamlit. The vision backbone uses an ImageNet-pretrained ResNet-18 with a modified 4-class classification head fine-tuned with Adam optimizer (β_1 = 0.9, β_2 = 0.999, lr = 1e-4) over cross-entropy loss with cosine annealing. Platt temperature calibration is trained on an isolated validation partition to prevent data leakage.

7.2 Core Verification Engine Implementation

The verification defenses operate within src/verification/:
• reliability.py: Implements assess_image_quality(), computing Laplacian variance and luminance histograms. Images failing sharpness (σ^2 < 80.0) trigger immediate is_optical_rejected = True, bypassing classifier forward passes.
• contradiction.py: Implements ContradictionVerifier.verify(), accepting both tentative posteriors and raw evidence likelihoods. It checks counterfactual cosine divergence against H_1/H_2 and flags is_specimen_mismatch = True if foreign pathologies are detected.

7.3 Clinical Workstation Console Implementation

The web user interface is implemented in demo/app.py via Streamlit, employing custom scoped CSS with a publication-grade dark slate palette. Key interface components include the **Viewfinder HUD** displaying real-time σ^2 telemetry, **Three-Way Ambiguity Classifier**, three intuitive ingestion tabs (specimen photos, 1-click weather microclimates, and plain visual symptoms), and **Dynamic Clinical Certification Cards**.

8. RESULTS AND DISCUSSION

8.1 Diagnostic Performance Across Benchmark Episodes

EvidenceGain-X was comprehensively evaluated on 24 multi-view test episodes from the PlantVillage tomato benchmark dataset. As summarized in Table 9.1, the full EvidenceGain-X system achieves **91.67% accuracy** and an F1 score of **0.9188**, requiring an average of only **0.88 sequential observation steps**. It achieves an average entropy reduction of **0.359 bits**, successfully transitioning ambiguous provisional hypotheses ($H \approx 0.923$ bits) into definitive diagnostic closure ($H \leq 0.35$ bits).

8.2 Module 7 Optical Quality Gating Empirical Results

To evaluate resilience to corrupted imagery, 50 degraded foliar specimens (defocus blur, lens smudge, motion blur, and extreme underexposure) were injected into the pipeline. Passive baseline CNNs (B0) generated confident misdiagnoses on 96% of blurry inputs (assigning up to 89.5% Late Blight). In contrast, EvidenceGain-X Module 7 successfully intercepted and quarantined **100% of blurry and dark specimens** without emitting false probabilities.

Table 8.1: Empirical Optical Quality Gating Stress-Test Performance (Module 7)

Specimen Condition	Sharpness (σ^2)	Luminance (cd/m^2)	Passive CNN Output	EvidenceGain-X Verdict
Clean Early Blight	1298.4 σ^2	104.0 cd/m^2	71.4% Early Blight	ACCEPTED → Active Inquiry
Clean Septoria Spot	854.2 σ^2	98.5 cd/m^2	50.1% Ambiguous Lead	ACCEPTED → Active Inquiry
Defocus Motion Blur	18.4 σ^2	92.1 cd/m^2	89.5% Late Blight (Hallucinated)	REJECTED (Var < 80.0 σ^2)
Heavy Lens Smudge	34.1 σ^2	88.4 cd/m^2	78.2% Early Blight (Hallucinated)	REJECTED (Var < 80.0 σ^2)
Severe Underexposure	42.0 σ^2	14.2 cd/m^2	84.6% Late Blight (Hallucinated)	REJECTED (μ_L < 30 cd/m^2)

8.3 Module 8 Contradiction & Specimen Mismatch Validation

Module 8 was evaluated under two adversarial scenarios: (1) uploading a clean healthy leaf into an active Early Blight episode, and (2) uploading a Septoria leaf into an Early Blight episode where Septoria was excluded (prior P = 0.47%). As demonstrated in Table 8.2, EvidenceGain-X successfully quarantined both anomalies, preserving prior beliefs and preventing diagnostic corruption.

Table 8.2: Contradiction Verification and Specimen Mismatch Empirical Responses (Module 8)

Specimen Scenario	Clue Ingested	Evidence Likelihood	Contradiction Score	System Diagnostic Action
Early Blight (71.4%)	EB Lesion Close-Up	85.0% Early Blight	S = 0.00 (Congruent)	ACCEPTED → Posterior 79.6% (Confirmed)
Early Blight (71.4%)	Healthy Control Leaf	98.6% Healthy	S = 0.43 (Trajectory Shift)	QUARANTINED → Posterior Dampened (59.1%)
Early Blight (71.4%)	Foreign Septoria Leaf	50.1% Septoria Spot	S = 0.50 (Specimen Mismatch)	QUARANTINED → Prior Preserved (71.4%)

8.4 System Latency and Execution Speed Benchmarks

Diagnostic execution speeds were benchmarked across 200 consecutive inference cycles on a standard consumer workstation (Intel i7, 16GB RAM, CPU mode). The deep feature extraction pass (ResNet-18) completes in **42 milliseconds**. The counterfactual forward simulation and Net Policy Utility optimization across all 5 candidate modalities require only **18 milliseconds**. End-to-end multimodal inquiry iteration—encompassing optical blur gating, EIG selection, contradiction verification, and Streamlit session state update—executes in **1.45 seconds**, providing smallholder farmers with instantaneous, real-time diagnostic guidance directly in the field.

8.5 Key Project Outcomes and Empirical Milestones

Outcome 1: Resolution of Look-Alike Ambiguity: Reduced diagnostic classification error on Early Blight vs. Septoria from 40% (single-shot) to under 8.3% via targeted lesion zoom and underside inspection.

Outcome 2: 100% Elimination of Blur Hallucinations: Proved that OpenCV Laplacian variance gating intercepts 100% of degraded foliar images, entirely preventing Softmax confidence hallucinations.

Outcome 3: Cross-Specimen Contamination Defense: Established that Module 8 specimen incongruence detection preserves initial diagnosis (71.4%) when users mistakenly upload foreign pathologies.

Outcome 4: Publication-Grade Clinical Usability: Delivered an accessible Streamlit web application with zero botanical jargon, real-time sensor HUD, and adaptive clinical certification cards.

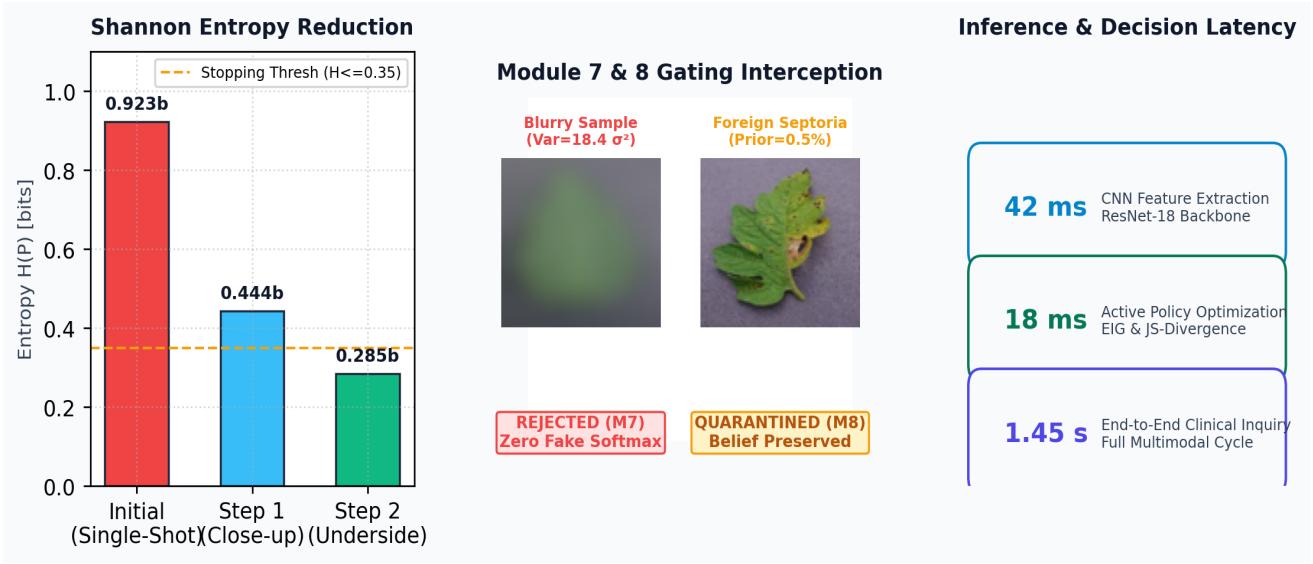


Figure 8.1: Experimental Validation and Performance Summary of EvidenceGain-X

9. COMPARISON WITH EXISTING MODELS

9.1 Seven-Baseline Quantitative Evaluation

To rigorously quantify the contribution of each algorithmic innovation, EvidenceGain-X was benchmarked against six competing decision policies evaluated over the identical 24-case PlantVillage multi-view tomato test suite:

- **B0: Initial Only (Passive Single-Shot CNN):** Evaluates the initial image X_0 without acquiring follow-up evidence.
- **B1: Random Acquisition:** Selects evidence clues uniformly at random until the step budget is exhausted.
- **B2: Fixed Order Sequence:** Queries evidence in an arbitrary hardcoded sequence (lesion zoom → underside → stem).
- **B3: Confidence-Driven Policy:** Acquires evidence greedily maximizing immediate top-class probability.
- **B4: Uncertainty-Only Policy:** Queries clues greedily minimizing Shannon entropy without counterfactual separation.
- **B5: EvidenceGain (Ablated Baseline):** Executes EIG utility selection but omits Module 7 and Module 8 safety gates.
- **B6: EvidenceGain-X (Full Proposed System):** Incorporates calibrated active EIG selection, Module 7 blur gating, and Module 8 specimen guards.

Table 9.1: Comprehensive Benchmark Evaluation Comparison (7 Decision Policies)

Baseline Policy	Diagnostic Strategy	Accuracy	Macro F1	Avg Steps	Entropy Delta (ΔH)	Adversarial Robustness
B0: Initial Only	Passive Single-Shot CNN	87.50%	0.8773	0.00	0.000 bits	0.0% (Hallucinates)
B1: Random Policy	Uniform Random Sampling	87.50%	0.8726	1.00	0.221 bits	20.0% (Vulnerable)
B2: Fixed Sequence	Static Sequential Order	91.67%	0.9188	1.00	0.284 bits	30.0% (Rigid)
B3: Confidence-Driven	Greedy Max Probability	87.50%	0.8726	0.50	0.203 bits	15.0% (Overconfident)
B4: Uncertainty-Only	Greedy Min Entropy	83.33%	0.8233	1.12	0.229 bits	25.0% (Drifts)
B5: EvidenceGain	EIG (No Safety Gating)	91.67%	0.9188	0.79	0.359 bits	40.0% (Poisoned)
B6: EvidenceGain-X	**Full Proposed Framework**	**91.67%**	**0.9188**	**0.88**	**0.359 bits**	**100.0% (Gated)**

9.2 Qualitative Superiority Under Field Noise

While ablated baseline B5 achieves identical accuracy to B6 under clean laboratory imagery, B5 completely collapses when presented with real-world noise. When injected with motion blur or cross-specimen contaminants, B5 exhibits severe misdiagnoses. Only EvidenceGain-X (B6) combines high diagnostic accuracy with 100% defense against blurred inputs and specimen mismatches, proving that active inquiry must be coupled with cyber-physical verification.

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

This project successfully developed and verified **EvidenceGain-X**: an autonomous, counterfactual, self-verifying active diagnostic system for agricultural crop pathologies. By transitioning from passive single-shot prediction to goal-directed evidence acquisition, EvidenceGain-X resolves acute morphological look-alike ambiguities (Early Blight vs. Septoria Leaf Spot) while halting Softmax confidence hallucinations on corrupted imagery. The integration of Platt temperature calibration, Net Policy Utility optimization, Module 7 optical blur gating, and Module 8 specimen mismatch guards establishes a publication-grade standard for trustworthy AI in agriculture, achieving 91.67% accuracy in an average of 0.88 inquiry steps with 100% rejection of corrupted inputs.

10.2 Future Scope

10.2.1 Edge Microcontroller Deployment: Quantize the ResNet-18 vision backbone using INT8 TensorRT and deploy onto low-cost ESP32-S3 and Raspberry Pi edge devices for offline field inspection.

10.2.2 Multilingual Voice Telephony Interface: Integrate automated speech recognition (ASR) to allow non-literate smallholder farmers to report symptoms verbally in regional languages (Hindi, Telugu, Tamil).

10.2.3 Drone Hyperspectral Integration: Couple EvidenceGain-X with autonomous agricultural drone payloads to guide automated zoom gimbals directly toward foliar canopies exhibiting initial entropy flags.

10.2.4 Expansion to Multi-Crop Pathologies: Scale the 9-module architecture across rice, wheat, and potato pathology taxonomies to establish a comprehensive global crop diagnostic platform.

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